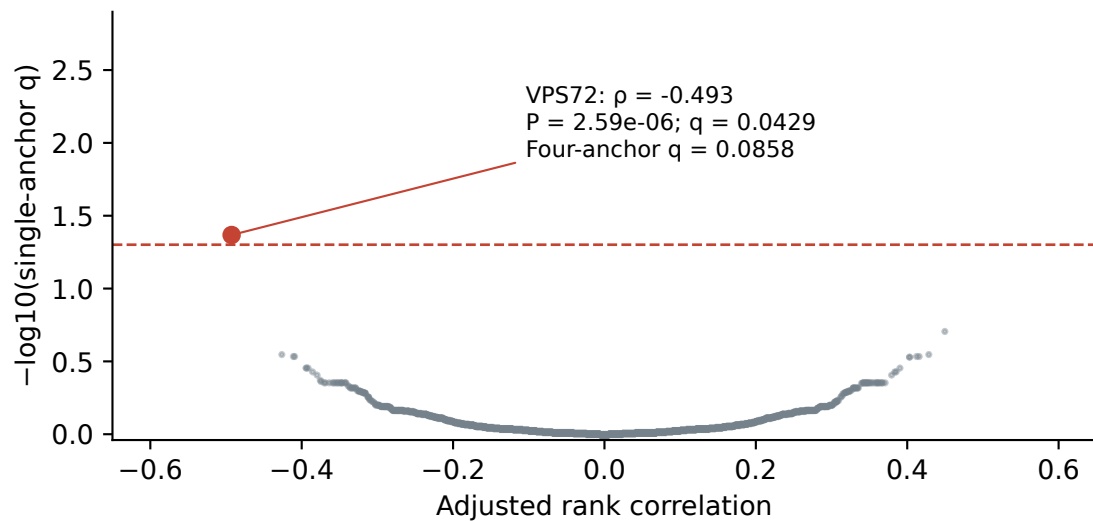
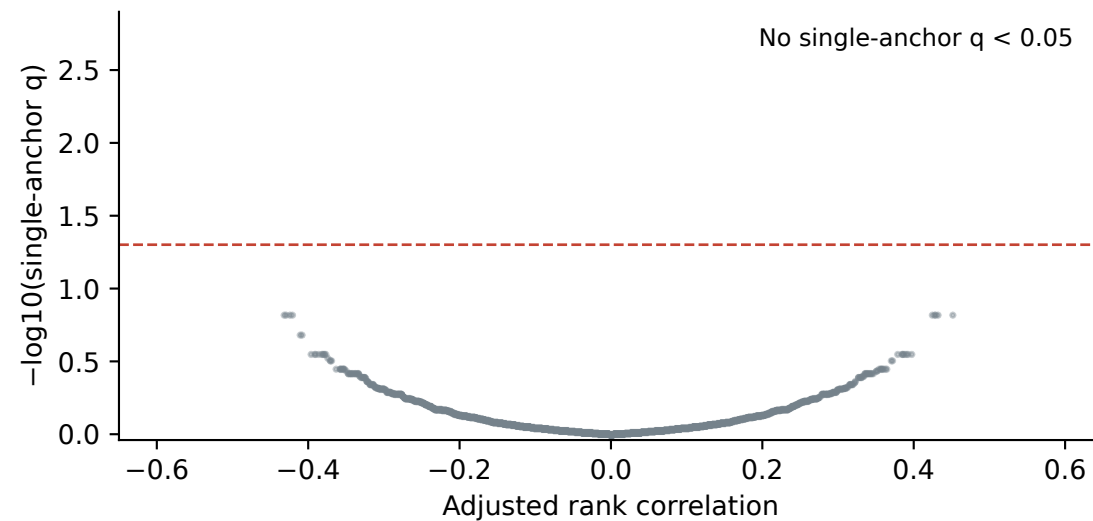


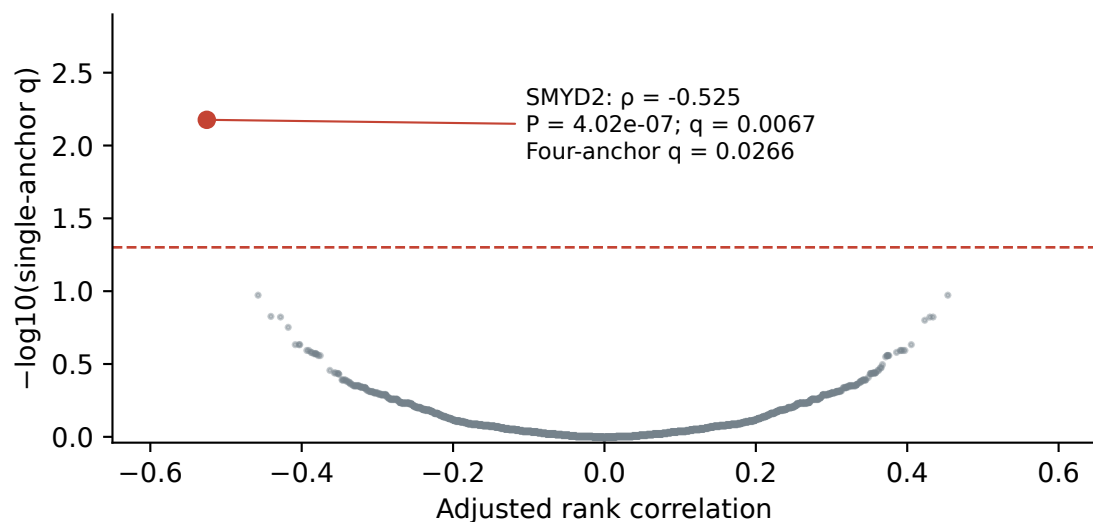
**A SCD adjusted organoid associations**



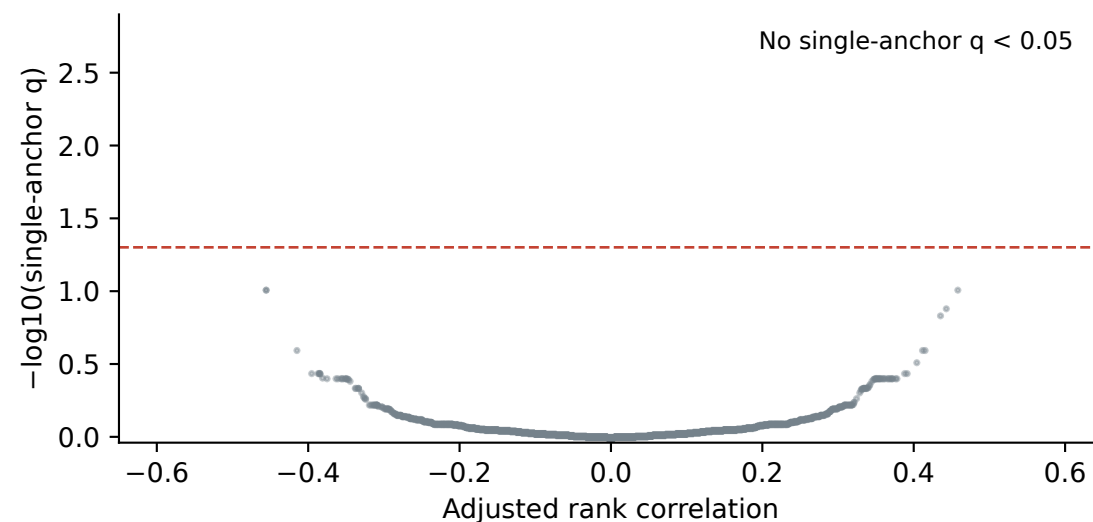
**B CH25H adjusted organoid associations**



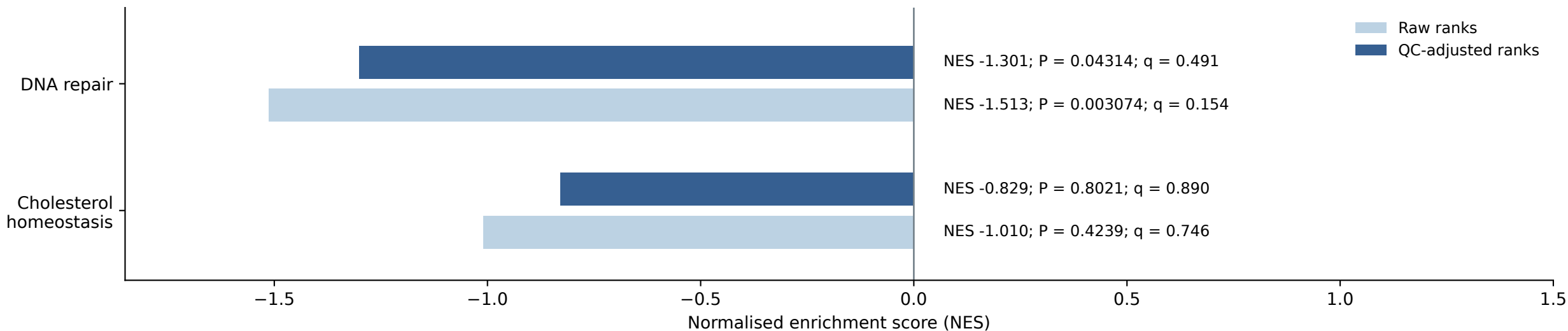
**C ATF3 adjusted organoid associations**



**D FADS2 adjusted organoid associations**



**E SCD Hallmark enrichment before and after QC adjustment**



A-D: 85 organoids; three QC covariates;  $df = 80$ ; 16,561 tests per anchor. E: neither full Hallmark ranking has  $q < 0.05$ .